

Zero Day Initiative — Finding Deserialization Bugs in the SolarWinds Platform

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Blog

Finding Deserialization Bugs in the SolarWinds Platform

** September 21, 2023

** Piotr Bazydło

It's been a while since I have written a blog post, please accept my sincerest apologies. This is because a lot of fun stuff that I've recently done is going to be presented during conferences.

Please treat this post as a small introduction to my upcoming [Hexacon 2023 talk](#) titled "Exploiting Hardened .NET Deserialization: New Exploitation Ideas and Abuse of Insecure Serialization". The entire talk and research was inspired by two small research projects, one of which focused on issues in SolarWinds deserialization.

In this blog post, I would like to present four old vulnerabilities that were fixed within the last year:

— [CVE-2022-38108](#) — [CVE-2022-36957](#) — [CVE-2022-36958](#) — [CVE-2022-36964](#)

A small part of the Hexacon talk will show how I have bypassed patches to some of these vulnerabilities. Right now, we will focus on the original issues.

CVE-2022-38108

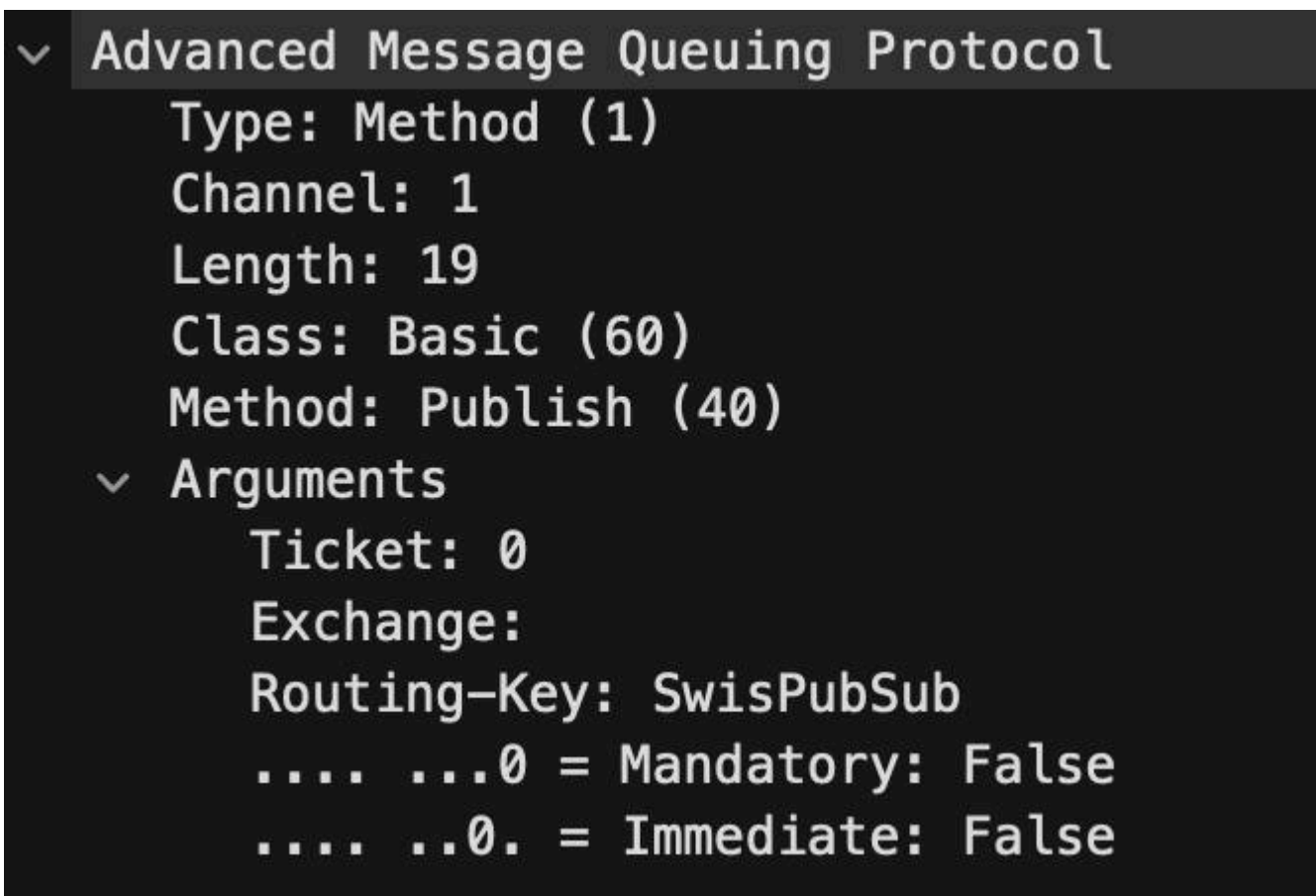
This vulnerability was already mentioned in this [blog post](#). Let me reintroduce it to you in more detail.

Several SolarWinds services communicate with each other through a RabbitMQ instance, which is accessible through port 5671/TCP. Credentials are required to access it. However:

— High-privileged users were able to extract those credentials through SolarWinds Orion Platform. — I later found [CVE-2023-33225](#), which allowed low-privileged users to extract those credentials.

This vulnerability targeted the SolarWinds Information Service. In order to deliver an AMQP message to the Information Service, the `Routing-Key` of the message must be set to `SwisPubSub`.

[View fullsize](#)



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Figure 1 - Routing-Key in AMQP message

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Now, let's verify how SolarWinds handles those messages! We can start with the `EasyNetQ.Consumer.HandleBasicDeliver` method:

At [1], the code retrieves the properties of the AMQP message. Those properties are controlled by the attacker who sends the message.

At [2] , it creates an execution context, containing both the AMQP message properties and the message body.

At [3] , it executes a task to consume the message.

This leads us to the `Consume` method:

At [1] , `EasyNetQ.DefaultMessageSerializationStrategy.DeserializeMessage` is called. It accepts the message properties and the message body as input. The interesting stuff happens here.

At [1] , we can see something really intriguing. A method named `Deserialize` is called and it returns an output of type `Type` . As an input, it accepts the `Type` property from the message. That's right - we can control `messageType` type through an AMQP message property!

At [2] , it calls `BytesToMessage` , which accepts both the attacker-controlled type and the message body as input.

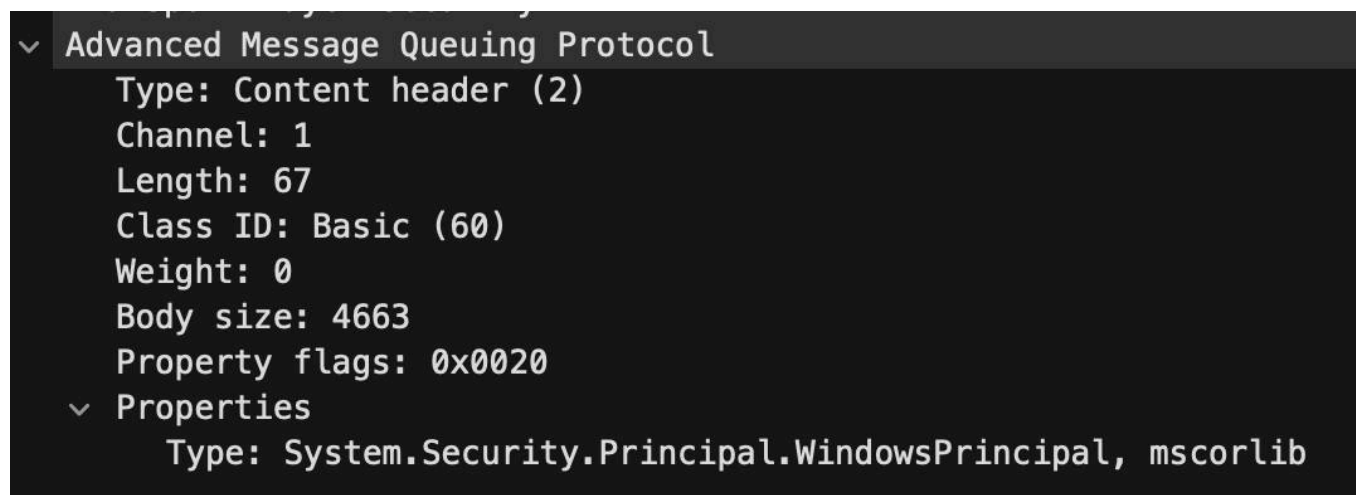
At [1] , the message body is decoded as a UTF-8 string. It is expected to contain JSON-formatted data.

At [2] , the deserialization is performed. We control both the target type and the serialized payload.

At [3] , it can be seen that the `TypeNameHandling` deserialization setting is set to `Auto` .

We have more than we need to achieve remote code execution here! To do that, we have to send an AMQP message with the `Type` property set to a dangerous type.

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Figure 2 - Deserialization Type control through AMQP properties

*

In the message body, we must deliver the corresponding JSON.NET gadget. I have used a simple `WindowsPrincipal` gadget from ysoserial.net, which is a bridge for the internally stored `BinaryFormatter` gadget. Upon the JSON deserialization, the RCE will be achieved through the underlying `BinaryFormatter` deserialization.

RCE achieved!

CVE-2022-36957

In the previous vulnerability, we were able to fully control the target deserialization type through the AMQP property. When I find such a vulnerability, I like to ask myself the following question: “What does a legitimate message look like?” I often check the types that are being deserialized during typical product operation. It sometimes leads to interesting findings.

I quickly realized that SolarWinds sends messages of one type only:

```
SolarWinds.MessageBus.Models.Indication
```

Let’s take a moment to analyze this type:

At [1] and [2], we can see two public members of type `SolarWinds.MessageBus.Models.PropertyBag`. The fun begins here.

At [1], you can see the definition of the class in question, `SolarWinds.MessageBus.Models.PropertyBag`.

At [2], a custom converter is registered for this class - `SolarWinds.MessageBus.Models.PropertyBagJsonConverter`. It implements the `ReadJson` method, which will be called during deserialization.

At [1], the code iterates over the JSON properties.

At [2], a JSON value is retrieved and casted to the `JObject` type.

At [3], a `Type` is retrieved on the basis of the value stored in the `t` key.

At [4], the object stored in the `v` key is deserialized, where we control the target deserialization type (again)!

You can see that we are again able to control the deserialization type! This type is delivered through the `t` JSON key and the serialized payload is delivered through the `v` key.

Let’s have a look at a fragment of a legitimate message:

We can take any property, for instance: `IndicationId`. Then, we need to:

- Set the value of the `t` key to the name of a malicious type.
- Put a malicious serialized payload in the value of the `v` key.

As the JSON deserialization settings are set to `TypeNameHandling.Auto`, it is enough to deliver something like this:

Now, let’s imagine that the first bug described above, CVE-2022-38108, got fixed by hardcoding of the target deserialization type to `SolarWinds.MessageBus.Models.Indication`. After all, this is the only legitimate type to be deserialized. That fix would not be enough, because `SolarWinds.MessageBus.Models.Indication` can be used to deliver an inner object, with an attacker-controlled type. We have a second RCE through control of the type here.

CVE-2022-36958

SolarWinds defines some inner methods/operations called “SWIS verbs”. Those verbs can be either: a) Invoked directly through the API. b) Invoked indirectly through the Orion Platform Web UI (Orion Platform invokes verbs internally).

There are several things that we need to know about SWIS verbs: • They are invoked using a payload within an XML structure. • They accept arguments of predefined types.

For instance, consider the `Orion.AgentManagement.Agent.Deploy` verb. It accepts 12 arguments. The following screenshot presents those arguments and their corresponding types.

View fullsize

EntityName	VerbName	Name	Type
Orion.AgentManagement.Agent	Deploy	pollingEngineId	System.Int32
Orion.AgentManagement.Agent	Deploy	agentName	System.String
Orion.AgentManagement.Agent	Deploy	hostname	System.String
Orion.AgentManagement.Agent	Deploy	ipAddress	System.String
Orion.AgentManagement.Agent	Deploy	machineUserName	System.String
Orion.AgentManagement.Agent	Deploy	machinePassword	System.String
Orion.AgentManagement.Agent	Deploy	additionalUsemame	System.String
Orion.AgentManagement.Agent	Deploy	additionalPassword	System.String
Orion.AgentManagement.Agent	Deploy	passwordIsPrivateKey	System.Boolean
Orion.AgentManagement.Agent	Deploy	privateKeyPassword	System.String
Orion.AgentManagement.Agent	Deploy	agentMode	System.Int32
Orion.AgentManagement.Agent	Deploy	installPackageFallbackId	System.String

*

Figure 3 - Arguments for Orion.AgentManagement.Agent.Deploy

*

The handling of arguments is performed by the method

`SolarWinds.InformationService.Verb.`

`VerbExecutorContext.UnpackageParameters(XmlElement[], Stream) :`

At [1], the `Type` is retrieved for the given verb argument.

At [2], a `DataContractSerializer` is initialized with the retrieved argument type.

At [3] and [4], the argument is deserialized.

We know that we are dealing with a `DataContractSerializer`. We cannot control the deserialization types though. My first thought was: I had already found some abusable `PropertyBag` classes. Maybe there are more to be found here?

It quickly turned out to be a good direction. There are multiple SWIS verbs that accept arguments of a type named `SolarWinds.InformationService.Addons.PropertyBag`. We can

provide arbitrary XML to be deserialized to an object of this type. Let's investigate!

At [1], the `ReadXml` method is defined. It will be called during deserialization.

At [2], the code iterates over the provided items.

At [3], the `key` element is retrieved. If present, the code continues.

At [4], the value of the `type` element is retrieved. One may safely assume where it leads.

At [5], the `value` element is retrieved.

At [6], the `Deserialize` method is called, and the data contained in both the `value` and `type` tags are provided as input.

At [7], the serialized payload and type name are passed to the `SolarWinds.InformationService.Serialization.SerializationHelper.Deserialize` method.

Again, both the type and the serialized payload are controlled by the attacker. Let's check this deserialization method.

At [1], the code checks if the provided type is cached.

If not, the type is retrieved from a string at [2].

At [3], the static `DeserializeFromStrippedXml` is called.

As you can see, the static `DeserializeFromStrippedXml` method retrieves a serializer object by calling `SerializationHelper.serializerCache.GetSerializer(type)`. Then, it calls the (non-static) `DeserializeFromStrippedXml(string)` method on the retrieved serializer object.

Let's see how the serializer is retrieved.

At [1], the code tries to retrieve the serializer from a cache. In case of a cache miss, it retrieves the serializer by calling `GetSerializerInternal` ([2]), so our investigation continues with `GetSerializerInternal`.

At [3], an `XmlTypeMapping` is retrieved on the basis of the attacker-controlled type. It does not implement any security measures. It is only used to retrieve some basic information about the given type.

At [4], an `XmlStrippedSerializer` object is initialized. Four arguments are supplied to the constructor:

- A new `XmlSerializer` instance, where the type of the serializer is controlled by the attacker(!).
- The `XsdElementName` of the target type, obtained from the `XmlTypeMapping`.
- The `Namespace` of the type, also obtained from the `XmlTypeMapping`.
- The type itself.

So far, we have two crucial facts:

- We are switching deserializers. The overall SWIS verb payload and arguments are deserialized with a `DataContractSerializer`. However, our `PropertyBag` object will eventually be deserialized with an `XmlSerializer`.
- We fully control the type provided to the `XmlSerializer` constructor, which is a key condition for exploitation.

It seems that we have it, another RCE through type control in deserialization. As `XmlSerializer` can be abused through the `ObjectDataProvider`, we can set the target deserialization type to the following:

```
System.Data.Services.Internal.ExpandedWrapper`2[[System.Web.UI.LosFormatter, System.Web,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a],
[System.Windows.Data.ObjectDataProvider, PresentationFramework, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=31bf3856ad364e35]], System.Data.Services,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e08
```

However, let's analyze the `XmlStrippedSerializer.DeserializeFromStrippedXml(String)` before celebrating.

Something unusual is happening here. At [1], a new XML string is being created. It has the following structure:

```
<XsdElementName xmlns='Namespace'>ATTACKER-XML</XsdElementName>
```

To sum up:

- The attacker's XML gets wrapped with a tag derived from the delivered type (see `GetSerializerInternal` method).
- Moreover, the retrieved `Namespace` is inserted into the `xmlns` attribute.

The attacker controls a major fragment of the final XML and controls the type. However, due to the custom XML wrapping, the `ysoserial.net` gadget will not work out of the box. The generated gadget looks like this:

The first tag is equal to `ExpandedWrapperOfLosFormatterObjectDataProvider`. This tag will be automatically generated by the `DeserializeFromStrippedXml` method, thus we need to remove it from the generated payload! When we do so, the following XML will be passed to the `XmlSerializer.Deserialize` method:

We still have a major issue here. Can you spot it?

When you compare both the original `ysoserial.net` gadget and our current gadget, one big difference can be spotted:

- The original gadget defines two namespaces in the root tag: `xsi` and `xsd`.
- The current gadget contains an empty `xmlns` attribute only.

The `ObjectInstance` tag relies on the `xsi` namespace. Consequently, deserialization will fail.

Luckily, the namespace does not have to be defined in the root tag specifically.

Accordingly, we can fix our gadget by defining both namespaces in the `ProjectedProperty0` tag. The final gadget is as follows:

In this way, we get a third RCE, where we fully control the target deserialization type!

Here is a fragment of the API request, where the malicious SWIS verb argument is defined:

CVE-2022-36964

Technically, this issue is identical to CVE-2022-36958. However, it exists in a different class that shares the same implementation of the `ReadXml` method. In this case, the vulnerable class is `SolarWinds.InformationService.Contract2.PropertyBag`.

An argument of this type is accepted by the `TestAlertingAction` SWIS verb, thus this issue is exploitable through the API.

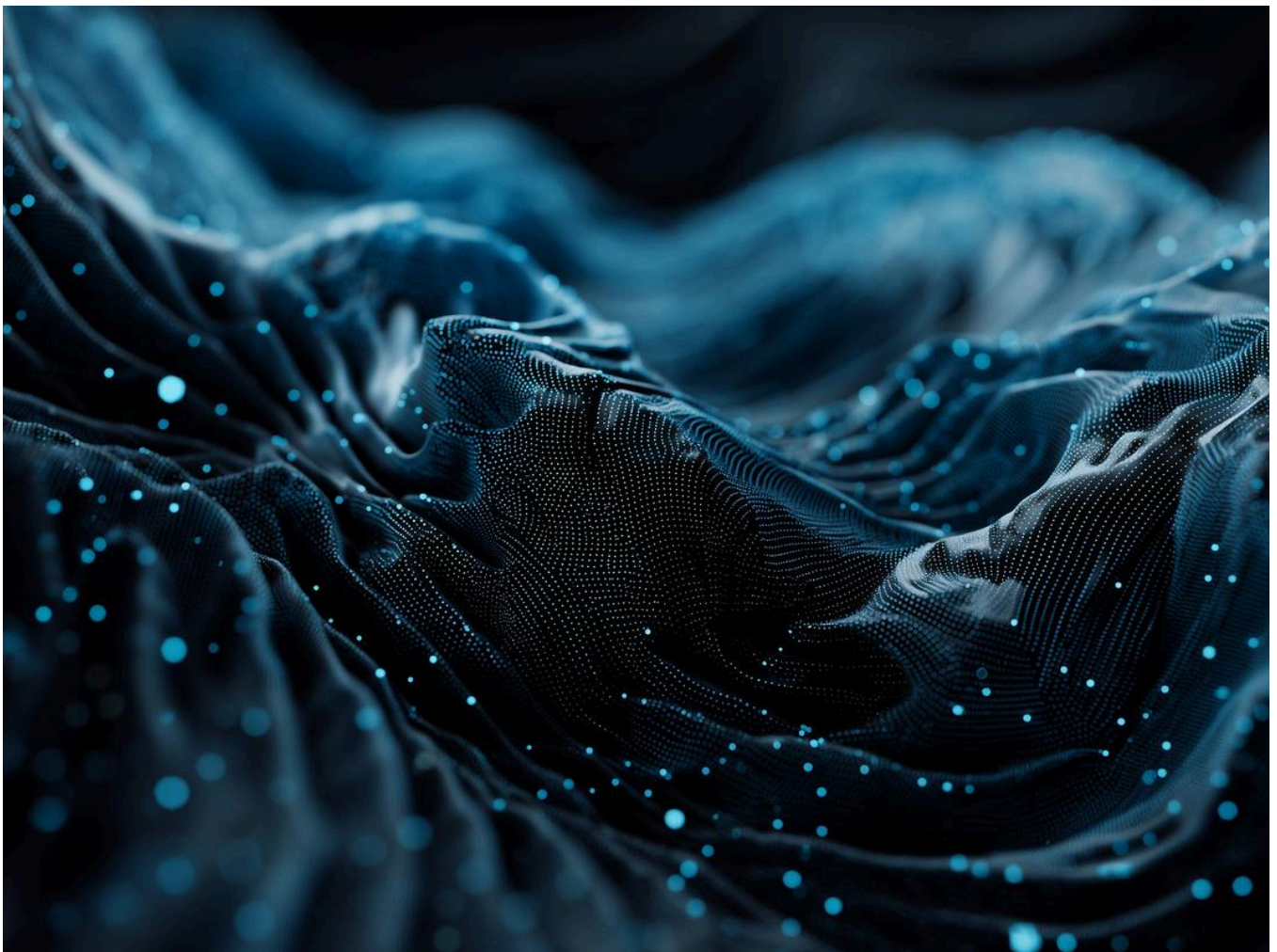
This class may appear familiar to some of you. I already abused that same class with JSON.NET deserialization in CVE-2021-31474. Almost one and a half years later, I realized that this class can be abused in a totally different way as well.

Summary

In this blog post, I have shown you four different deserialization vulnerabilities in SolarWinds where the attacker could control the type of the deserialized object. One of them was particularly interesting, because `DataContractSerializer` could be used to ultimately reach `XmlSerializer`. During my Hexacon 2023 talk, I will show you some of the patches applied to the described issues and I will show you how I have bypassed them by using custom deserialization gadgets. These patch bypasses have also been patched by SolarWinds, but the discussion will show how hunting deserialization bugs can lead to some fun discoveries.

I hope you liked this writeup. Until my next post, you can follow me [@chudypb](#) and follow the team on [Twitter](#), [Mastodon](#), [LinkedIn](#), or [Instagram](#) for the latest in exploit techniques and security patches.

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